ID token Vs access token

ID Token and Access Token មានទំរង់ ដូច JWT (jwt = json web token) វាមានលក្ខណៈ រូបរាង ដូចគ្នា មាន Format ដូច json ដែល។

A JWT is just the shape of the card. An ID token and an access token are different cards that may use that same shape.

1- ID Token: "Who are you?" ប្រាប់ថាអ្នកជាអ្នកណា
2- Access Token = "What are you allowed to enter?" យើងដែលចូលមកនឹងមានសិទ្ធអ្វីខ្លះ

Simple analogy

Imagine you go to an office building.

1. ID Token = "Who are you?"
This is like your **employee ID card**.

It tells:

- your name
- your employee number
- that you really signed in

So the app looks at the ID token and says:

> "Okay, this is Camtech. He has logged in."

Its job is to prove **your identity**.

2. Access Token = "What are you allowed to enter?"
This is like a **door pass**.

It tells:

- which room or service you can access
- what permissions you have
- how long the pass is valid

So the API looks at the access token and says:

> "Okay, this request is allowed to read user profile data."

Its job is to prove **permission**.

- **ID token** = for identity
- **Access token** = for permission
- **JWT** = the format used to carry the information

Very simple version

When you log in:

- **ID token** answers: **"Who am I?"**
- **Access token** answers: **"What can I use?"**

Super easy analogy

```
Plain text

ID Token     = your name card
Access Token = your permission pass
JWT          = the envelope/card format
```

Example:

Sample ID Token

This token is for **identity**.

```
Plain text

eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9
.
eyJpc3MiOiJodHRwczovL2xvZ2luLmV4YW1wbGUuY29tIiwiYXVkIjoiY2xpZW50LTEyMyIsInN1YiI6InVzZ
.
SIGNATURE
```

Decoded payload of that ID token

```
JSON

{
  "iss": "https://login.example.com",
  "aud": "client-123",
  "sub": "user-789",
  "email": "camtech@example.com",
  "name": "Camtech KH",
  "iat": 1700000000,
  "exp": 1700003600
}
```

What it means

- `iss` = who created the token
- `aud` = which app it is for
- `sub` = user id
- `email` and `name` = user info
- `iat` = issued time
- `exp` = expiry time

So this is basically saying:

> "This user is Camtech KH, and this token was issued for app client-123."

Sample Access Token

This token is for **API permission**.

Plain text


```
eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9
.
eyJpc3MiOiJodHRwczovL2xvZ2luLmV4YW1wbGUuY29tIiwiYXVkIjoiaHR0cHM6Ly9hcGkuZXhhbXBsZS5jb
.
SIGNATURE
```


Decoded payload of that access token

JSON

```
{
  "iss": "https://login.example.com",
  "aud": "https://api.example.com",
  "sub": "user-789",
  "scope": "read:profile write:settings",
  "iat": 1700000000,
  "exp": 1700003600
}
```

What it means

- `aud` = this token is for the **API**
- `scope` = what actions are allowed

So this is basically saying:

> "User user-789 is allowed to call the API and can read profile and write settings."